Technology Report and Presentation

Design and Creative Technologies

Torrens University, Australia

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Subject Code: CCF 501

Subject Name: Cloud Computing Fundamentals

Assessment No.: 1

Title of Assessment: Technology Report and Presentation

Lecturer: Dr. Dyvia Leekha

Date: Mar 2026

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Cloud Computing for ABC Enterprise: Key Contributions to Business Automation

1. Executive Summary

Cloud computing has become a core business utility - email, storage, and smart devices all run on someone else's infrastructure, delivered on demand (Nishimura, 2022). For ABC Enterprise from our case study, this model translated directly into results: start-up IT costs dropped by ~80% and the platform absorbed a 10x customer surge within a month, without a matching increase in headcount (Eliaçık, 2022; McHaney, 2021). This report outlines three key benefits of cloud adoption, the main challenges to manage, and a practical recommendation for service model, deployment model, cost model, and cloud provider. Where ABC needs a provider beyond "XYZ", AWS is recommended for its maturity, global reach, and automation-ready managed services (Amazon Web Services, n.d.).

2. Benefits of Cloud Computing vs. Traditional IT Infrastructure

Traditional IT means owning servers, cooling, and the staff to keep it running (McHaney, 2021). For a high-growth start-up like ABC, that model is a strategic handicap. Cloud flips it: instead of buying capacity, ABC rents capability, aligned to the NIST characteristics of on-demand self-service, measured service, rapid elasticity, and resource pooling (Mell & Grance, 2011).

2.1. Cost Efficiency and Pay-as-you-grow Model

Cloud shifts spend from CAPEX to OPEX - no upfront servers, no idle hardware costs (Eliaçık, 2022). ABC's ~80% reduction in start-up IT costs is the "measured service" characteristic in action: pay for compute-hours, storage GB-months, and data transfer actually consumed (Mell & Grance, 2011). As McHaney (2021) notes, operational overhead - personnel, training, upgrades, and security - often exceeds hardware cost over time, making pay-as-you-go a structural advantage. Beyond the bill, standard tasks like backups, patching, and scaling can be codified and automated, reducing human toil across the delivery pipeline (Carpenter, 2020).

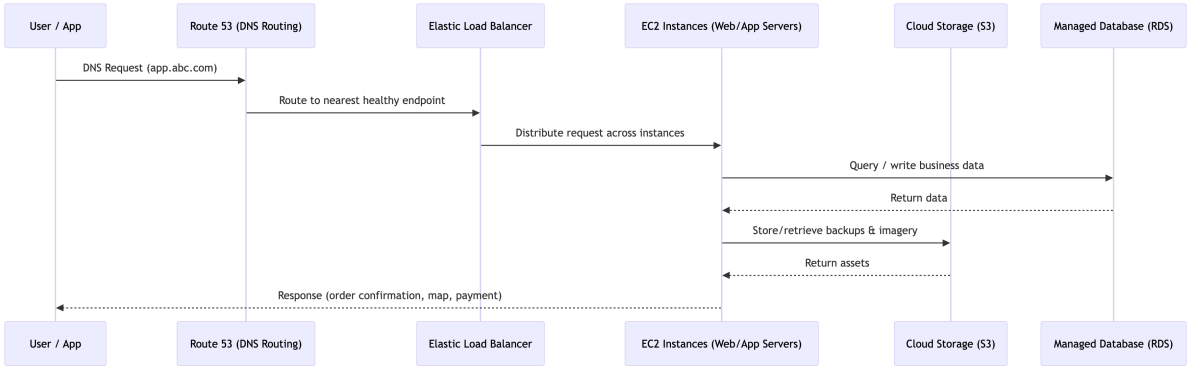

Figure 1 – ABC Enterprise cloud traffic flow - from user request through Route 53, load balancing, compute, storage, and database layers

2.2. Rapid Scalability for Business Growth

A 10x customer surge in a single month exposes the core weakness of on-premises infrastructure: procurement lead times mean hardware arrives after the opportunity has passed (McHaney, 2021). Cloud's rapid elasticity lets ABC scale compute up during

campaigns or city launches and scale down when demand normalizes - automatically, not reactively (Mell & Grance, 2011). Eliaçık (2022) highlights that this scalability protects against outages from traffic spikes - critical for an app handling delivery, taxi bookings, and payments.

2.3. Reduced IT Management Overhead

In on-premises environments, more customers mean more infrastructure and more staff to maintain it (McHaney, 2021). Cloud breaks that linear relationship. Through resource pooling, providers consolidate physical resources across tenants — Manvi and Shyam (2021) describe this as partitioning and isolation that allows multiple workloads to share hardware. ABC gains automated maintenance and resilient architectures that would be expensive to replicate in-house (Naved et al., 2022).

3. Challenges of Cloud Migration and Mitigation Strategies

Cloud adoption is not risk-free. Three challenges are most relevant for ABC (Eliaçık, 2022):

1. **Security and Privacy:** ABC handles payments and customer PII. Security is the top concern for cloud adopters - 90% of security professionals cite it as a challenge, alongside governance (71%) and compliance (68%) (Manvi & Shyam, 2021). Mitigation: least privilege IAM, mandatory MFA, encryption at rest/in transit, and continuous alerting - the shared responsibility model clarifies the provider/customer division (Shore, 2020).

2. **Cost Volatility:** Pay-as-you-go can spiral without guardrails - overprovisioned instances and excessive egress generate surprise bills (Bittok, 2022). **Mitigation:** FinOps habits from day one: budget alerts, resource tagging, rightsizing, and reserved pricing for stable workloads.
3. **Vendor lock-in and Skills gap:** Deeper managed-service adoption makes provider migration expensive (Eliaçık, 2022). Mitigation: prioritize portability (containers, standard databases) and invest in targeted upskilling (McHaney, 2021).

4. Recommended Cloud Models for ABC Enterprise

Cloud service models sit on a control-versus-responsibility spectrum (IBM, n.d.-a; Linthicum, 2021; McHaney, 2021). IaaS provides full compute flexibility; PaaS abstracts infrastructure so the team can focus on development; SaaS offers limited customisation - less suited to a start-up that must differentiate its platform.

Deployment models vary in elasticity and cost. Community cloud does not apply to ABC's consumer use case. Private cloud sacrifices elasticity; hybrid introduces complexity that exceeds ABC's current maturity (Manvi & Shyam, 2021). Public cloud is the most strategically aligned option: IBM (n.d.-b) reports IaaS workloads experience 60% fewer security incidents than traditional data centers (see Table 1).

Deployment Model	Cost	Elasticity	ABC Fit
Public Cloud	Low (OPEX only)	High / auto-scales	Recommended
Private Cloud	High (CAPEX + ops staff)	Limited / fixed capacity	Over-engineered for startup
Hybrid Cloud	Medium - dual infra	Moderate / complex to manage	Premature / exceeds current maturity

Table 1 – Deployment model trade-offs for ABC Enterprise

Recommendation: A blended IaaS/PaaS approach - IaaS for compute flexibility, PaaS for managed databases and serverless functions - preserves rapid elasticity and measured service (Mell & Grance, 2011). A Virtual Private Cloud (VPC) can address future network isolation needs (IBM, n.d.-c).

5. Recommended Cost Model

Cloud providers offer three levers:

1. Pay-as-you-go (maximum flexibility, highest unit price),
2. Reserved/committed pricing (discounts for baseline commitments),
3. Spot/preemptible (deep discounts for interruption-tolerant workloads) (Bittok, 2022; Amazon Web Services, n.d.-b).

Recommendation: A hybrid cost model - reserved capacity for stable customer-facing tiers (web/app, databases), pay-as-you-go autoscaling for demand spikes, and spot instances for background jobs and analytics pipelines. TCO modelling should include not just cloud fees but

also migration cost, security tooling, and engineering effort (Bittok, 2022). As Bittok (2022) notes, cloud adoption is rarely about the cheapest bill - it's about better ROI: less downtime, faster launches, and automation that avoids linear headcount growth.

6. Cloud Provider Recommendation: AWS

Azure suits Microsoft-aligned enterprises; GCP leads in analytics. AWS best fits ABC: Route 53 already anchors the described architecture, and AWS offers the broadest automation-ready managed-service catalogue (Nishimura, 2022; Amazon Web Services, n.d.-a) (see Table 2).

Provider	Ecosystem Fit	Load Balancing	Serverless	ABC Alignment
AWS	**Broadest managed-service catalogue**	**ELB: native Route 53 integration**	**Lambda: event-driven, zero idle cost**	**Best fit: Route 53 already in stack**
Azure	Microsofot / enterprise-aligned	Application Gateway / extra config needed	Azure functions: strong but separate ecosystem	No existing Azure signals in ABC
GCP	Analytics and ML-first	Cloud Load Balancing: GKE-oriented	Cloud Run / Functions - container-first	No analytics-heavy workloads in ABC now

Table 2 – Cloud provider comparison for ABC Enterprise.

1) **Elastic Load Balancing:** Automates traffic routing and health-checks across EC2 instances - critical for ABC's latency-sensitive services. Integrates natively with Route 53 and Auto Scaling (Amazon Web Services, n.d.-a).
2) **Auto Scaling:** Provisions or terminates EC2 instances on demand signals, directly implementing NIST's rapid elasticity (Mell & Grance, 2011). The 10x surge required no manual intervention - capacity becomes policy-driven, not operator-driven (Amazon Web Services, n.d.-a).

3) **AWS Lambda:** Event-driven compute for ABC's workflow automation: order placed → delivery assigned; payment confirmed → restaurant notified. Scales to zero when idle, charging only per invocation (Amazon Web Services, n.d.-a).

7. Conclusion

Cloud computing is the right strategic move for ABC: measured service, rapid elasticity, and resource pooling (Mell & Grance, 2011) translate directly into automation outcomes - scaling without procurement delays, zero headcount growth, faster time-to-market. Recommended stack: public cloud, blended IaaS/PaaS, hybrid cost model, AWS. Managed well, ABC's infrastructure stops being a bottleneck and becomes a competitive advantage (Eliaçık, 2022).

8. Appendices

8.1. Appendix A – Cloud vs Traditional IT Comparison Table

Dimension	Traditional IT	Cloud (ABC Case)
Cost model	CAPEX heavy	OPEX elasticity
Scaling	Hardware procurement	Auto Scaling
Deployment speed	Weeks/months	Minutes
Automation	Manual ops	Policy-driven

Table 3 – Cloud vs Traditional IT Comparison.

8.2. Appendix B – High-Level Architecture Diagram

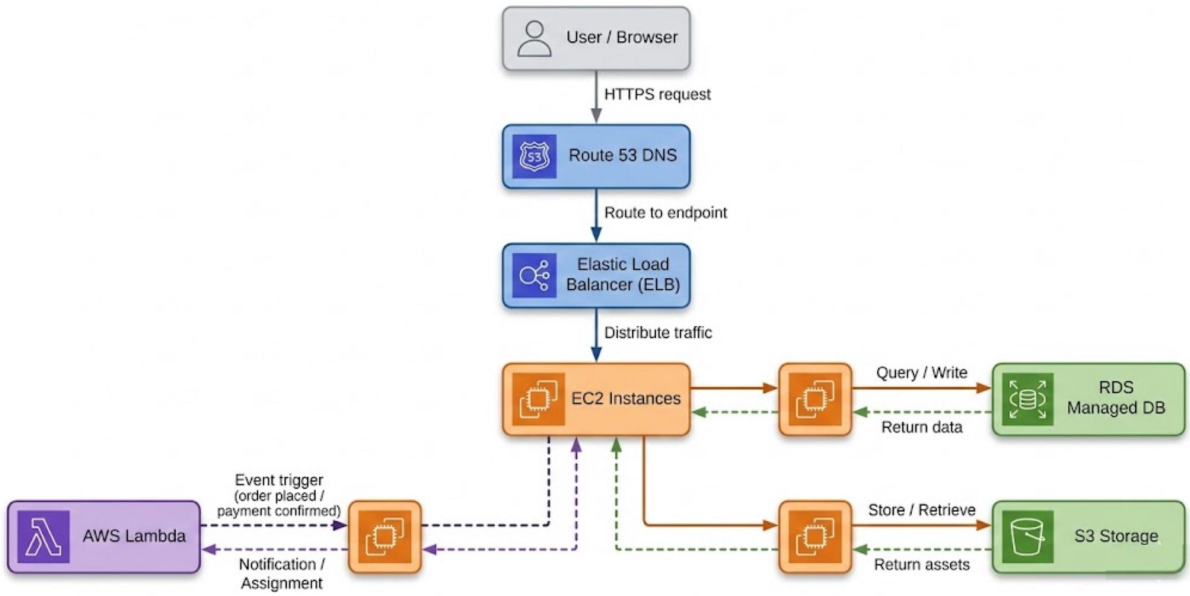

Figure 2 – ABC Enterprise high level cloud architecture on AWS.

8.3. Appendix C – Cost Model Breakdown Example

Cost Category	On-Premises	Cloud
Hardware	High upfront	None
Maintenance	Internal team	Managed
Scaling	Hardware purchase	Auto scaling
Downtime cost	High	Reduced

Table 4 – Cost Model Breakdown.

8.4. Appendix D – Shared Responsibility Model Diagram

Cloud Provider secures	ABC secures
Physical infra	IAM
Networking	Applications
Hypervisor	Data governance

Table 5 – Shared Responsibility Model.

8.5. Appendix E – NIST Five Essential Characteristics Mapped to ABC

NIST Essential Characteristic	ABC Enterprise Mapping
On-Demand Self-Service	Dev team provisions EC2/RDS via console — no vendor call needed
Broad Network Access	App accessible via mobile, browser, API across delivery + taxi regions
Resource Pooling	ABC shares AWS physical hardware; logically isolated via VPC
Rapid Elasticity	10× customer surge absorbed automatically via Auto Scaling
Measured Service	~80% cost reduction — pay only for compute-hours, GB-months consumed

Table 6 – NIST essential cloud characteristics mapped to ABC Enterprise (Mell & Grance, 2011).

8.6. Appendix F – Cloud Resources Provisioned (What we are hiring)

AWS Service	Role	Baseline Config	Auto-Scale Ceiling
EC2 (web/app tier)	Serve API requests for all three verticals (Food, Taxi, Payments)	2x t3.medium (2vCPU, 4GB RAM)	20x c5.xlarge (4vCPU, 8GB RAM)
Auto Scaling	Scale EC2 fleet up/down on demand signals	Policy-driven (CloudWatch triggers)	Absorbed 10x surge with zero manual intervention
Elastic Load Balancer (ELB)	Distribute inbound traffic across EC2 fleet	Always-on (hourly-charge)	Scales transparently – no manual config
RDS (PostgreSQL)	Structured data: orders, rides, payments, users	Db.r5.large (2vCPU, 16GB) - Multi-AZ	Read replicas added on demand
S3	Receipt storage, media assets, backups	Pay-per-GB – no provisioned minimum	Unlimited (object storage)
AWS Lambda	Event-driven workflows: order assignment, payment notifications, driver dispatch	128MB / 3s timeout per function	1,000 concurrent executions (default limit, raisable)
Route 53	DNS routing and health checks	Always-on (per-query billing)	Globally redundant by default
VPC	Network isolation – multi-tenant segmentation per vertical	Single VPC with subnet per workload tier	Peering and private endpoints as needed
CloudFront	CDN – low-latency delivery of static assets to distributed users	Global edge network (always-on)	Scales to any volume; invalidation on deploy
CloudWatch	Monitoring, alerting, autoscale triggers	Always-on (free tier + paid metrics)	Retains 15 months of metrics
Cost Explorer	FinOps – cost tracking, rightsizing, usage analysis	Always-on (no provisioning needed)	Validates ~80% cost reduction claim

AWS WAF + Shield Standard	Security layer – DdoS mitigation, traffic filtering	Included (Shield Standard free)	Shield Advanced available for high-volume attacks

Table 7 – AWS services provisioned for ABC Enterprise — baseline configuration and scaling ceiling.

8.7. Appendix G – Glossary of Terms

Term	Meaning
CAPEX	Capital Expenditure
OPEX	Operational Expenditure
IAM	Identity and Access Management
TCO	Total Cost of Ownership
SOA	Service-Oriented Architecture
PaaS	Platform as a Service
IaaS	Infrastructure as a Service
SaaS	Software as a Service
VPC	Virtual Private Cloud
ELB	Elastic Load Balancer
EC2	Elastic Compute Cloud
RDS	Relational Database Service
Route 53	AWS DNS Service
Auto Scaling	AWS service for automatic scaling of compute resources
Shared Responsibility	Cloud security model defining provider vs customer security obligations
Measured Service	Resource usage is monitored and billed accordingly
Rapid Elasticity	Allows resources to scale up/down automatically based on demand

Table 8 – Glossary of Terms

End of Appendix Section

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